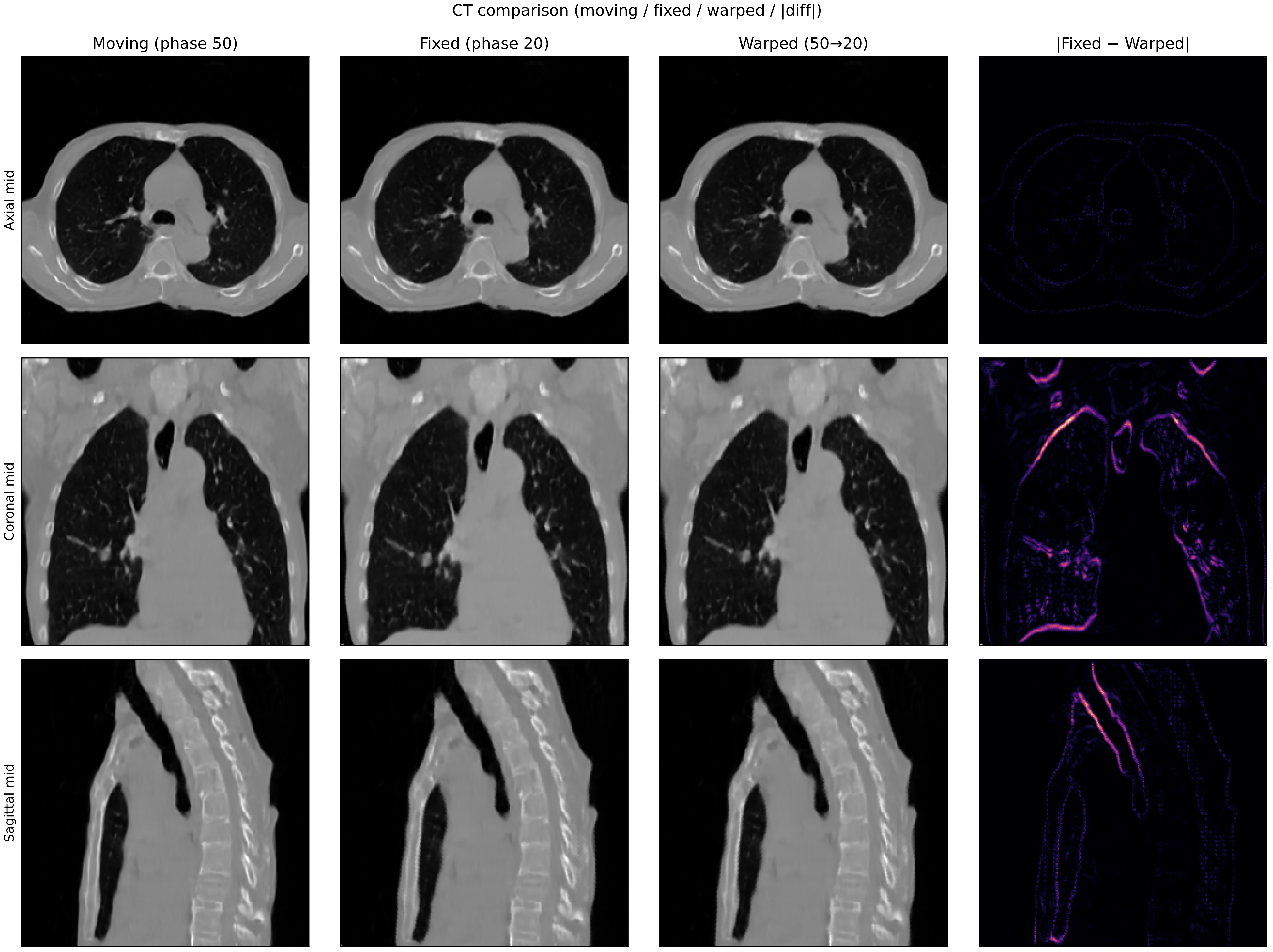
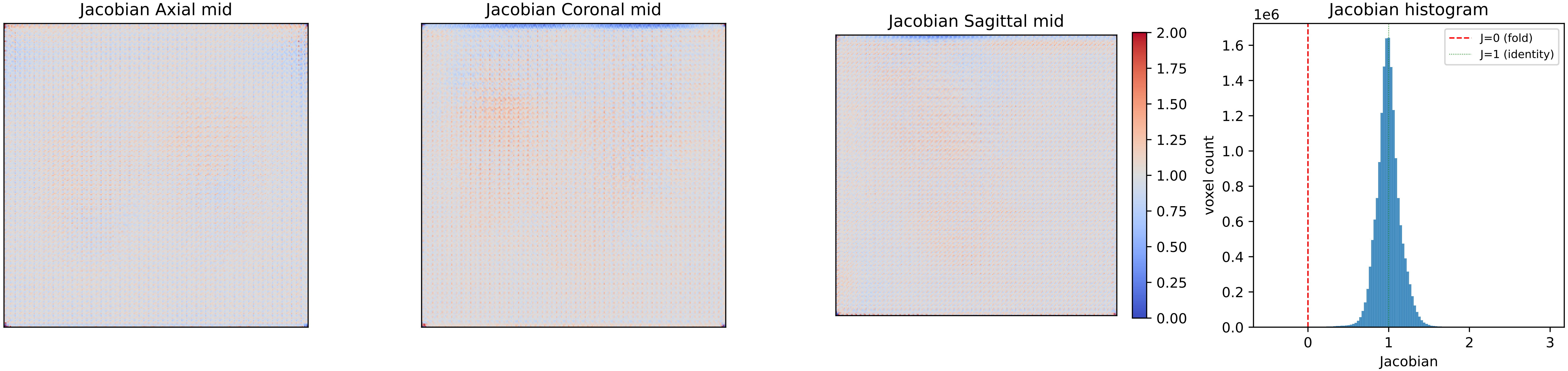
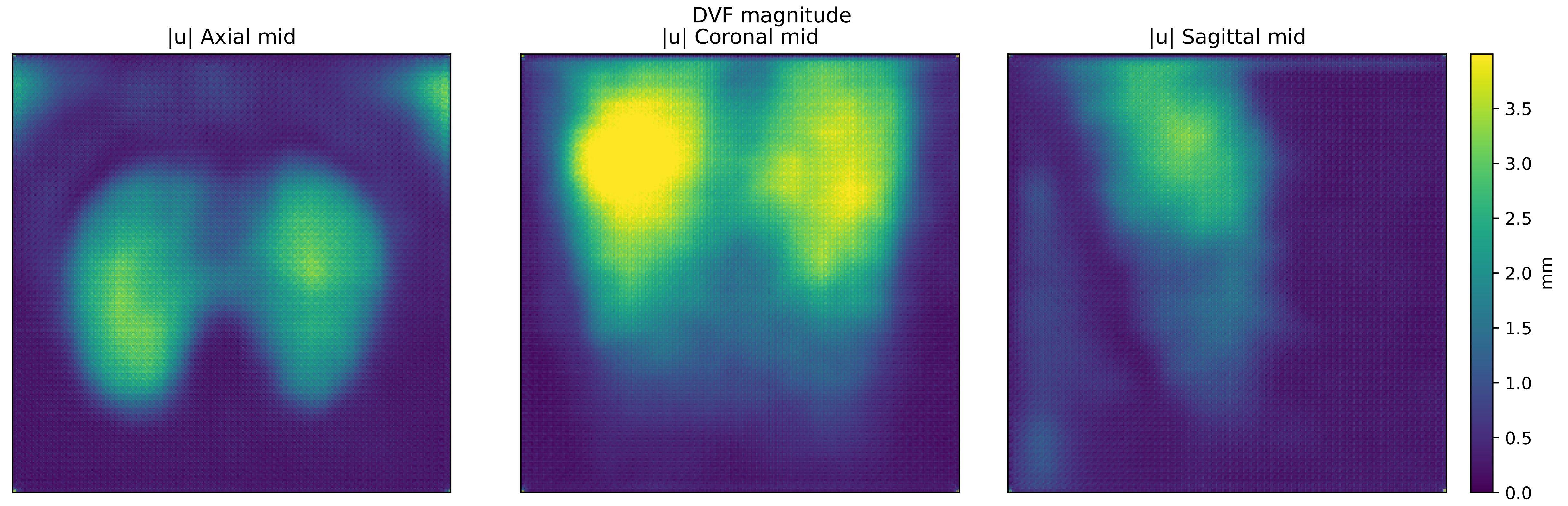
DVF QA metrics — phase 20

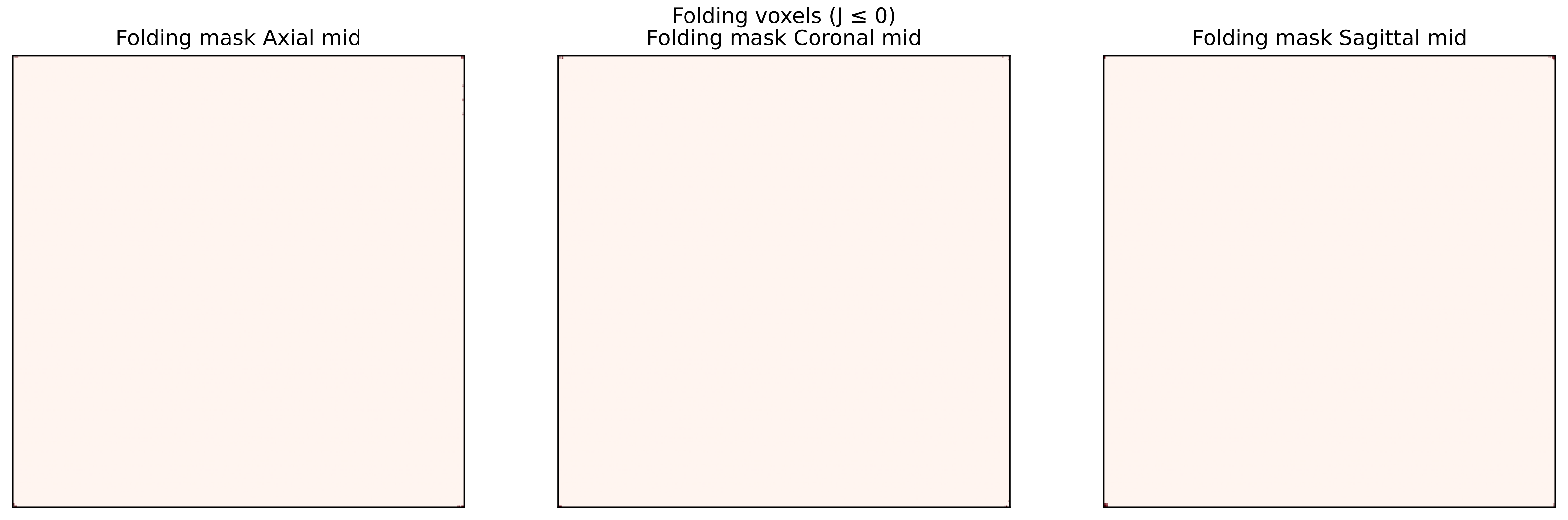
Metric	Value
qa_status	FAIL
jac_min	-0.0843592
jac_p01	0.683266
jac_p05	0.780284
jac_median	0.992332
jac_mean	0.997578
jac_p95	1.23974
jac_p99	1.35336
jac_max	2.7449
fraction_jac_le_0	6.4e-08
fraction_jac_lt_0_2	2.5856e-05
fraction_jac_gt_5	0
folding_component_count	1
largest_folding_component_ml	0.001
dvf_magnitude_p95_mm	3.10596
dvf_magnitude_max_mm	6.12319
gradient_norm_mean	0.218528
bending_energy_mean	0.046137
warped_vs_fixed_mse	3724.39
warped_vs_fixed_mae	20.1405
warped_vs_fixed_ncc	0.992116
our_warp_vs_precomputed_mse	3724.39
our_warp_vs_precomputed_mae	20.1405
our_warp_vs_precomputed_ncc	0.992116


CT comparison (moving / fixed / warped / |diff|)
Moving (phase 50)
Fixed (phase 20)
Warped (50→20)
|Fixed − Warped|
Axial mid
Coronal mid
Sagittal mid

Jacobian determinant + histogram




DVF magnitude
|u| Axial mid
|u| Coronal mid
|u| Sagittal mid
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
mm


Folding voxels (J ≤ 0)
Folding mask Axial mid
Folding mask Coronal mid
Folding mask Sagittal mid